

Research paper

Size doesn't matter: Assessing the trustworthiness of large language models in medical contexts: A focus on epidural information retrieval

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ABSTRACT

Background: Since the release of ChatGPT, numerous LLMs have emerged, providing easy access to information without the need for technical expertise. However, relying on these systems can influence important life decisions, such as the choice to use epidural analgesia during childbirth.

Epidural analgesia is widely regarded as the “gold standard” for pain relief during childbirth. However, limited access to anaesthesiologists and gaps in knowledge may prompt individuals to seek information from unverified sources, including AI systems. Misinformation in this area can discourage the use of effective analgesia, highlighting the need to assess the accuracy of LLM-generated content.

Objective: To evaluate the reliability of LLM-generated information regarding epidural analgesia.

Methods: We posed 10 standardized questions about epidural analgesia to 12 LLMs, each question reformulated 10 times in both Spanish and English, resulting in 2400 responses. Two anaesthesiologists were involved in the assessment process. One expert performed the initial ratings, while the second independently verified the evaluations assessing the outputs using an extended SERVQUAL framework.

Results: ChatGPT performed best, followed by Gemini 2. Medium-sized models, such as Phi-3 and OpenChat, outperformed several larger models like Llama-2 or Llama-3, challenging the notion that “bigger is better” and offering potential advantages in low-resource settings (e.g., Phi-3 outperformed Llama-2 with an average increase of 81% across all metrics). Specialized models did not show superior performance. Except for ChatGPT, English responses were generally more reliable than Spanish, with some Spanish outputs incoherent. ChatGPT also exhibited the least variability between responses.

Conclusions: Despite promising performance, LLMs display limitations in medical contexts. Collaboration between national and international medical societies is crucial to develop evidence-based resources to guide LLM training and improve information trustworthiness.

1. Introduction

During labour, there are numerous sensations expressed uniquely and differently by each woman. The pain of childbirth is generally experienced as a very painful and high-intensity sensation, influenced by numerous factors, including physical, psychological, and social ones [1, 2]. Childbirth pain causes emotional suffering, psychological problems, and a series of maternal and fetal physiological alterations [3].

Today, there are numerous analgesic alternatives available [4], with neuraxial analgesia – and specifically epidural analgesia – being considered the “gold standard” for pain relief during childbirth, according to Brogly, Manrique and Guasch's Care Protocols of the Obstetric Anaesthesia Section of SEDAR [5], and different studies focusing on pain management during labour [4,6,7]. Epidural analgesia is a technique performed by an anaesthesiologist, in which a catheter is inserted into the epidural space at the lumbar level, through which appropriate medication is administered. It is an effective technique throughout all

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stages of labour and has a high safety profile. The anaesthesiologist must individualize pain management, considering the unique characteristics of each patient as well as their preferences. For this, they must know the risk stratification factors for increased pain, including fear or heightened anxiety [5]. Therefore, the information available prior to labour is fundamental and can affect the experience of this process and the perceived pain. Despite a large number of women going through childbirth over their lifetimes, knowledge about analgesic methods – and more specifically, epidurals – is insufficient [8–12].

The decision to use epidural analgesia during childbirth is based on factors such as knowledge, experience, attachment to tradition, social position, and trust in the country's healthcare system [8,9]. At times, women may reject epidural analgesia due to misinformation and fear of potential side effects they believe may occur [10,13].

Access to anaesthesiologists for resolving doubts is often limited, prompting many women to seek information from non-anaesthesiologist healthcare professionals, family, friends, or online sources [8–12]. Individuals increasingly obtain health-related information online [14–16], and pregnant women are among the most active users, with up to 92% seeking prenatal information on the internet [17]. Yet online content – particularly on platforms such as TikTok and YouTube – often contains misleading or inaccurate health information [15,16,18]. This includes epidural analgesia, where recent analyses show substantial inaccuracies that may shape women's decisions [19]. Up to 45% of women who decline epidural analgesia base their choice on fear of non-existent side effects, sometimes influenced by media reports [10]. For these reasons, epidural analgesia counselling provides a clinically meaningful and high-stakes test case for evaluating AI-generated answers or large language models (LLMs) trustworthiness: it is common, emotionally charged, and involves preference-sensitive decisions that depend on balanced communication of risks and benefits. As patients increasingly consult LLMs for medical guidance [20], epidural counselling offers a relevant and rigorous context for assessing whether LLM-generated information is sufficiently trustworthy in this domain and safe for patient decision-making.

AI has the potential to reduce medical errors by optimizing processes, improve and complement image interpretation, increase patient knowledge about their own data, and promote their health [21–24]. In anaesthesia, AI is currently primarily used for monitoring the depth of anaesthesia, controlling anaesthesia, predicting events and risks, guidance in ultrasound, pain management, and operating room logistics [25]. Almost all doctors in the future will use AI technology [21].

All of this has generated as much excitement as concern about the potential risks involved [26]. Some of the most common criticisms of generative AI are its inappropriate responses, lack of impartiality in both political and racial contexts, and the invention of facts and references. OpenAI has already worked on these issues, achieving significant improvements in GPT-4 [26]. However, LLMs are still in early development stages and require significant improvements [26]. They aspire to produce human-like responses, as exemplified by the Turing test [27]. Despite the enthusiasm, concerns around generative AI persist [26]. Common criticisms include biased or inappropriate outputs, hallucinated facts, and lack of transparency.

Within the medical field, interest in generative AI is growing. A 2023 survey of 1000 UK general practitioners revealed that 20% had already used AI for clinical documentation and diagnostic suggestions [28]. These developments highlight the necessity of educating both clinicians and patients on responsible AI use in healthcare.

Given the general public's tendency to trust AI-generated outputs, there is a real risk that flawed responses could influence health decisions. At present, the dangers of failed AI outputs outweigh those of direct clinician-patient interactions [21]. To safely integrate AI into clinical workflows, more rigorous research, clinical validation, and peer-reviewed studies are needed [21,29]. Medical professionals must also help define quality standards, enforce oversight, and raise awareness of LLMs' current limitations [30,31].

To explore these issues, our study undertakes a rigorous analysis of a diverse range of LLMs – both public and private – across different sizes and training purposes. Our primary objective is to assess the trustworthiness and decision safety of medical information generated by LLMs for patients, as opposed to focusing solely on general factual accuracy, with particular attention to how deficiencies, ambiguities, or inaccuracies may influence real clinical decisions. We include general-purpose models such as ChatGPT, Gemini, Gemma, Phi-2, Phi-3, OpenChat, Llama-2, and Llama-3; open-source variants like Llama-2-Unc. and TinyLlama; and medically specialized models such as MedLlama and Meditron [32]. These models are evaluated for their performance in providing trustworthy medical information, with a specific focus on epidural analgesia.

Our evaluation framework extends the SERVQUAL model from Choi et al. [33], assessing seven key dimensions: tangibility, reliability, responsiveness, certainty, empathy, comprehensibility, and concordance with an anaesthesiologist's perspective. We formulate 10 core questions about epidural analgesia, each rewritten 10 times, in 2 languages, and submit them to 12 different LLMs, generating a total of 2400 responses. Evaluations are performed in both English and Spanish to assess multilingual performance. The secondary aims of our evaluation are: (1) to assess multilingual performance (English vs. Spanish) in order to understand how language influences LLM knowledge; (2) to evaluate the impact of model size on response quality and determine whether size is truly relevant; (3) to compare general-purpose and medically specialized LLMs to examine whether specialized models outperform general-purpose ones when addressing the general public; and, (4) to distinguish between trustworthy answers and those that could misinform patients and potentially alter their decision-making.

In this work, we place a strong emphasis on patient-centred evaluation, distinguishing our study from clinician-focused or purely technical benchmarks by prioritizing safety, clarity, and the potential impact of LLM-based generated information on patient decision-making. A key methodological contribution is our adaptation of the SERVQUAL framework to assess trustworthiness in patient-facing medical AI, which we highlight as a central innovation well-aligned with the needs of real-world patient communication. We also underscore the importance of multilingual assessment – here conducted in both English and Spanish – as a critical component for addressing global health equity and for understanding how language-dependent variability in LLM performance may shape accessibility and fairness. Finally, we clarify that our evaluation focuses on consistency with clinical guidance and safety-relevant reliability rather than model size or raw capability, reflecting the priorities that matter most in patient-facing medical contexts.

The results indicate that ChatGPT is the top-performing model, with even better performance in Spanish. Interestingly, medium-sized open models like Phi-3 and OpenChat outperformed larger models. Furthermore, domain-specific models like MedLlama and Meditron did not offer superior results, suggesting that general-purpose models may be equally or more effective in certain clinical contexts. The main contributions of this paper are:

- We present the first large-scale manual evaluation (2400 answers) comparing 12 LLMs on epidural analgesia for childbirth information. This rigorous human-in-the-loop assessment extending the SERVQUAL models establishes a high-quality benchmark of 16,000 manual evaluations for medical and general LLMs in real-world patient education scenarios.
- Our results show that medium-sized models (Phi-3, OpenChat) can match or outperform large models, challenging the “bigger is better” assumption. This is especially valuable for low-resource settings where computational constraints limit the use of very large models.
- We demonstrate the impact of language on LLM performance, with ChatGPT improving in Spanish, unlike other models, and highlight the critical risks of relying on some medical-specialized or open models in clinical settings such as MedLlama or Meditron.

- We publish the evaluation dataset and code (on GitHub) to enable transparency, reproducibility, and further research in AI safety and trustworthiness in sensitive medical contexts.¹

The extended SERVQUAL framework can be applied to other patient-facing medical AI contexts — such as oncology counselling, vaccination advice, and other areas where patients rely on clear, trustworthy, and safety-critical information.

2. Background

Generative AI creates text, audio, and images that appear human-made, producing an illusion of intelligence based on vast datasets [34]. On November 30, 2022, ChatGPT – a highly capable, general-purpose generative AI model – became publicly available and rapidly popular. Its release sparked unprecedented interest, amassing 100 million users in two months [35] and over 200 million weekly active users by December 2024 [36].

Following ChatGPT's success, other large language models (LLMs) emerged [26]. One notable example is Meta's Llama-2 [37], which gained traction after its weights were released for public use. This enabled the development of derivative models — some fine-tuned on domain-specific datasets [38], others retrained using Llama-2-generated data [39]. Today, both general-purpose models (e.g., ChatGPT, Llama-2, Gemini) and domain-specific ones (e.g., MedLlama, Meditron) are available, with the latter tailored for medical applications.

Generative AI can assist with document creation, image generation, or music composition through natural language interactions, or “prompts”. These systems typically require no technical background and are accessible through free or low-cost subscriptions (e.g., €20/month for ChatGPT, €21.99 for Gemini).

Despite demonstrated productivity improvements in various fields [40], the adoption of these technologies in medicine remains controversial, primarily due to concerns over data privacy and reliability. For instance, GPT-4 was evaluated in a study involving 10 hand-related medical cases, scoring an average of 4.6 out of 5 as judged by orthopaedic hand surgeons [41]. Furthermore, LLMs like ChatGPT have passed medical exams such as the USMLE and Spain's MIR [42,43], and even provided more empathetic and informative responses to medical queries than some professionals [44]. Studies using echocardiograms have shown that cardiologists were unable to distinguish between interpretations made by AI and those made by experienced specialists [45]. Similarly, for chest X-rays, AI has demonstrated non-inferiority to radiologists in detecting active tuberculosis [46]. Yet, other research found ChatGPT matched doctors' final diagnoses in 82.1% of evaluated cases [47]. Conversely, a preprint examining responses about juvenile idiopathic scoliosis found deficiencies in empathy, contextualization, and appropriate language, underlining the need for continued refinement [48]. Generally, studies comparing model performance (e.g., ChatGPT-3.5 vs. Gemini) have not evaluated a broader range of LLMs [49].

Nonetheless, serious limitations persist. In one study simulating clinical interviews, GPT-4 was asked to generate medical notes. On average, 23.6 errors were found per case – mostly omissions, followed by additions and factual inaccuracies – with variation across repeated generations [50]. These inconsistencies highlight the risks of clinical deployment.

It is essential to educate users on responsible use—particularly the importance of not inputting confidential data. These models cannot “unlearn” once trained, and the data may persist indefinitely [51]. Reproducibility is another concern, as LLM responses often vary for the same query. Although potentially useful for patients, their vague, non-specific answers may limit utility for specialists [52].

While generative AI can mimic understanding, it fundamentally operates on statistical patterns without true comprehension. This leads to reasoning errors, factual inaccuracies, dangerous recommendations, hallucinated references, and even simple arithmetic mistakes [53]. The text they produce often appears polished and credible, making errors hard to detect—even by experts [30].

To specialize models for specific tasks, a method called fine-tuning is used, training them on targeted data. However, this and the generative process require substantial computing power—typically involving GPUs. Reducing model size is one solution, as seen with projects like TinyLlama, though it comes at the cost of lower output quality. In medicine, where accuracy is paramount, this trade-off poses a significant concern.

Because many LLMs are under development or proprietary, evaluating their true capabilities and limitations remains challenging. This work aims to contribute a critical methodology for assessing generative AI quality in a specific medical domain: epidural analgesia during childbirth. Key considerations include understanding model parameters, quantifying uncertainty, and evaluating response quality through repeated querying. The ultimate goal is to test whether these systems can reliably inform the general public about epidurals as a pain management option during labour.

3. Methodology

Our methodology, summarized in Fig. 1, aims to measure trust in different large language models (LLMs) within the context of medical advice. To that end, we focus on advice related to *epidural analgesia*—a common method used for pain relief during childbirth. The methodology is divided into three main parts.

First, a medical expert establishes a set of common questions to be asked of the LLMs. These questions vary in difficulty and reflect typical concerns of the general public. These concerns were identified based on questions raised by women in previous studies contraindications, secondary effects, short term and long-term negative consequences of epidural analgesia, risk of paralysis [9,10], as well as on frequently asked questions encountered by the authors. Each question is also rephrased in multiple ways to assess the models' ability to understand and respond to differently worded queries.

Second, we select a range of large language models to evaluate. Once the models are chosen, we collect and assess their responses.

Third, we use a set of evaluation metrics – extending the SERVQUAL model [33] – to quantify the quality of the answers. These metrics not only assess trustworthiness, but also the reliability of the models in a multilingual context and the consistency of their responses.

Although this study focuses specifically on epidural analgesia, the methodology is designed to be generalizable to other medical topics. Ultimately, our goal is to understand the trustworthiness of large language models when providing medical information to the general public.

3.1. Questions about epidural analgesia

To evaluate the different LLMs, we have selected 10 questions in public-facing, conceptual queries related to epidural procedures. The 10 questions were carefully selected and paraphrased to simulate diverse real-world patient inquiries, allowing us to assess model consistency, multilingual reliability, and alignment with clinical standards.

1. *Does an epidural increase the chances of having a C-section?* This question was considered controversial many years ago. Earlier studies in this field suggested a possible link [54] whereas more recent research indicates otherwise [5].
2. *What are the most common complications of an epidural during labour?* This is a straightforward and commonly asked question.

¹ <https://github.com/hdg7/epidural>

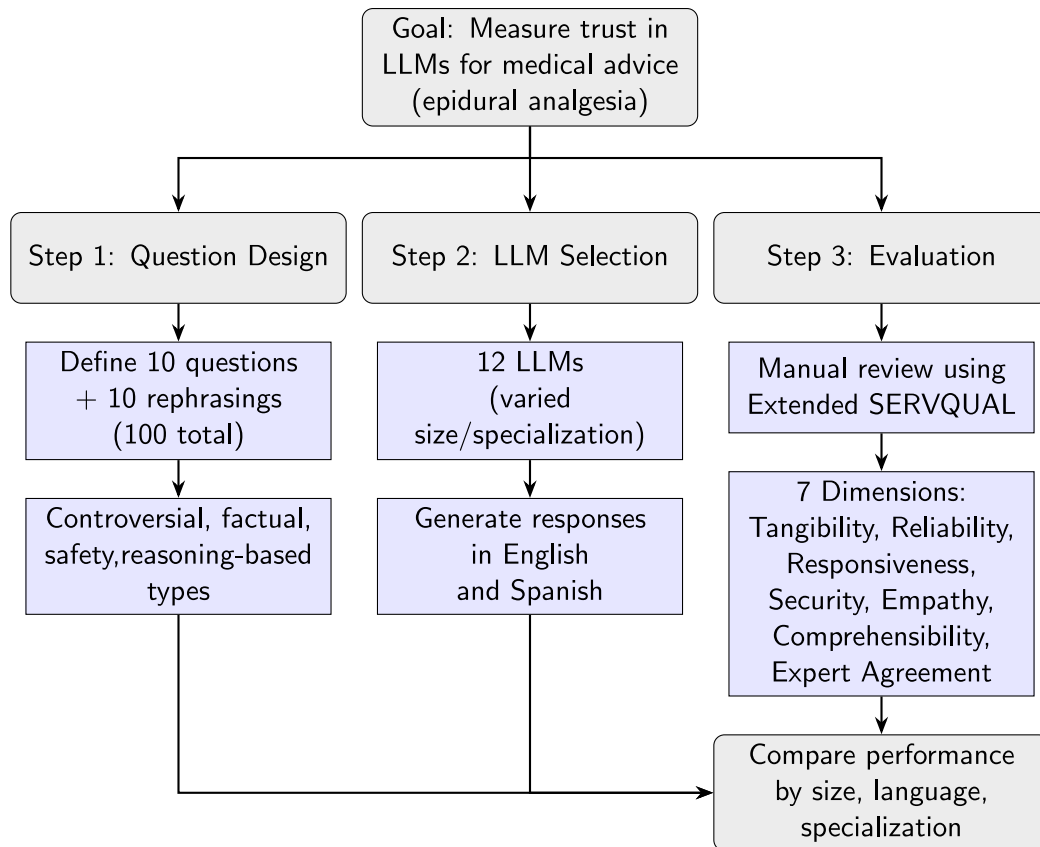


Fig. 1. General steps of our LLMs evaluation methodology extending the SERVQUAL model.

3. *What are the long-term risks of an epidural during labour?* This question may be slightly confusing, as it focuses on long-term rather than short-term consequences.
4. *What are the contraindications for getting an epidural during labour?* This is a direct medical question.
5. *Who administers the epidural during labour?* This is a straightforward factual question.
6. *What could happen if I move while having the epidural during labour?* This question requires reasoning and is therefore considered more difficult.
7. *Will I be able to push during labour if I get an epidural?* This is a common and direct question.
8. *Can I become paralysed after getting an epidural during labour?* This is a relatively simple question, though it touches on common fears.
9. *When can I get an epidural during labour?* This is a direct and factual question.
10. *Is getting an epidural during labour safe for the baby?* This is a direct question addressing common safety concerns.

To assess the consistency of the LLMs, each question has been rephrased in multiple ways. Specifically, we have rewritten and reformulated each one to generate ten variations per question. This results in a total of 100 different questions posed to each LLM. The study focuses on the quality of information and its alignment with expert guidance, rather than on clinical decision-making in complex scenarios. The objective is to formulate questions that closely resemble those real patients might ask in their everyday lives and to evaluate the LLMs' performance in this context.

3.2. Selection of large language models

The population to be studied includes various general-purpose open-source LLMs, as well as models specifically trained for medical applications. The models under evaluation are listed in Table 1.

We were also interested in evaluating artificial intelligence models across different configurations. Specifically, we focused on three categories: **small-size** models, **medium-size** models, and **large-size** models. We define small-size models as those with approximately 1 to 2 billion parameters, medium-size models as those with around 7 billion parameters, and large-size models as those with roughly 70 billion parameters. With respect to the rest of the parameters, we left the default values at the time of the study, we are specified in Table 1.

Models such as ChatGPT and Gemini, whose exact configurations are not publicly disclosed, are of particular interest because they allow us to compare performance against smaller, open-source models. These models were accessed through the public chat interface, performing the questions directly to them to simulate the common behaviour of the general public. The versions during the study were GPT-4 and Gemini1.0-Pro. These proprietary models are believed to be significantly larger, with estimated parameter counts ranging from 540 billion for Gemini to over 1.7 trillion for ChatGPT [55]. Our aim is to identify models that can deliver comparable results using fewer computational resources.

We have also ensured representation from a diverse range of developers, including Meta, OpenAI, Google, Microsoft, and community created models published in Hugging Face. Additionally, we included models specifically trained for medical tasks – such as MedLlama and Meditron – to evaluate how they compare to general-purpose models and to assess their reliability for public-facing medical information.

Table 1

Comparison of AI Language Models, including their default parameters during the experiments in April 2024.

Model	Company	Size	Access	Params.	T, k, p
ChatGPT	OpenAI	Very Large	Private	1 T	1.0, –, 1.0 ^a
Gemini	Google	Very Large	Private	540 B	1.0, 40, 0.95 ^b
Llama-2	Meta	Large	Public	70 B	1.0, 50, 1.0 ^c
Llama-3	Meta	Large	Public	70 B	1.0, 50, 1.0 ^c
TinyLlama	Community	Small	Public	1.1 B	1.0, 50, 1.0 ^c
Gemma	Google	Medium	Public	2 B	1.0, 50, 1.0 ^c
Llama-2-Unc.	Community	Large	Public	70 B	1.0, 50, 1.0 ^c
Phi-2	Microsoft	Medium	Public	2.7 B	1.0, 50, 1.0 ^c
Phi-3	Microsoft	Medium	Public	2.7 B	1.0, 50, 1.0 ^c
OpenChat	Community	Medium	Public	7 B	0.8, 40, 0.9 ^c
MedLlama	Community	Large	Public	70 B	1.0, 50, 1.0 ^c
Meditron	Community	Medium	Public	7 B	1.0, 50, 1.0 ^c

^a Indicates the information from OpenAI documentation.^b From Google Vertex AI documentation.^c Indicates that is employing HuggingFace, Transformers or Ollama default versions.

3.3. Evaluation metrics

Each question has been evaluated using a set of metrics.

In order to rate the answers as accurately as possible, the latest guidelines were first reviewed to ensure that the information was up to date. Additionally, the answers that were to be marked as acceptable were discussed by two experts. Once both anaesthesiologists agreed on the accepted answers, they were manually verified by one to ensure consistency and alignment with established clinical guidelines. Both raters, the first (MBB) and third authors (MJVL) of this work, have extensive experience in anaesthesiology and relied on the latest procedural manuals to objectively assess the responses. MJVL is the head of the anaesthesiology department, and MBB is a staff member with eight years of experience in the field.

To ensure maximum precision, each response was evaluated using an extended version of the SERVQUAL model [33], which focuses on five core dimensions: tangibility, reliability, responsiveness, security and certainty, and empathy. We added two additional dimensions to the original model: **comprehensibility**, as the investigators identified varying levels of readability in the responses; and **agreement with the anaesthesiology expert**, to reflect alignment with expert medical opinion (summarized in Fig. 2).

Additionally, for responses generated in languages other than English, we examined how well the language used aligned with the expected form based on the input question.

Each dimension was rated on an ordinal scale, offering a detailed assessment of each model's performance.

The first dimension, **tangibility**, assesses the structural coherence of a response. A well-organized response with clear construction receives the highest rating (5), while a slightly less robust but logical structure is rated (4). A neutral structure is scored (3), a relatively poor structure lacking clarity is rated (2), and a poorly structured response receives the lowest score (1).

Reliability measures the trustworthiness of the content. A highly reliable and accurate response is rated (5), followed by generally reliable answers (4). A neutral rating (3) indicates acceptable accuracy. Responses with questionable reliability are rated (2), and those with serious inaccuracies receive the lowest score (1).

The **responsiveness** dimension evaluates the response's alignment with current medical knowledge. The highest scores are given to responses containing up-to-date information (5) or reflecting the latest guidelines (4). Neutral responses receive a (3), while outdated or poorly informed answers are rated (2) or (1), respectively.

Security and certainty measure how safely the information can be used from a medical perspective. Responses that can be confidently used as a reference are rated (4), while those that are sufficiently accurate for general trust receive a (3). Moderate reliability is rated (2), and unsafe or misleading responses receive the lowest score (1).

Empathy assesses the model's ability to understand the context of the question. Full comprehension is rated (4), partial understanding receives a (3), basic or neutral responses are rated (2), and those that fail to grasp the question's intent are rated (1).

Comprehensibility evaluates how easy the response is to understand. Very clear and easy-to-read responses are rated (5), relatively easy ones receive a (4), while average clarity is rated (3). Difficult-to-understand responses receive a (2), and very confusing ones a (1). This dimension was introduced to address specific gaps observed during the preliminary analysis. Comprehensibility captures the clarity and readability of LLM-generated responses, which is critical for patient-facing communication.

Finally, **agreement with the anaesthesiology expert** measures how well the response aligns with expert opinion. Full agreement is rated (4), general agreement receives a (3), insufficient responses are rated (2), and those containing serious errors or misinformation are given the lowest score (1). It assesses whether the model's output is consistent with clinical standards and expert consensus. Both dimensions (comprehensibility and agreement with anaesthesiology expert) were developed in consultation with domain experts and are grounded in the need to evaluate not only factual correctness but also the practical utility of responses in real-world medical contexts.

This evaluation framework provides a rigorous and systematic method for comparing the performance of different language models, offering insights into their strengths and limitations in the context of medical advice.

4. Results

This section presents the main research questions addressed in this study, focusing on evaluating the quality, robustness, and multilingual performance of various language models in the context of epidural analgesia. Our methodology involves expert-generated questions, expert-guided answer validation, topic-specific analysis, and metric-based evaluations across different large language models from different contexts (Table 1).

RQ1. How good are the different large language models with respect to the specific metrics used to evaluate them?

The primary objective of this question is to determine how well large language models perform in generating high-quality answers to clinical questions about epidural analgesia. We evaluate the models using a set of quality metrics, including *tangibility*, *comprehensiveness*, *empathy*, *reliability*, *responsiveness*, and *agreement with expert opinions*. Each model is given the same set of questions, and their responses are assessed by a medical expert based on these criteria. This evaluation helps establish a baseline understanding of each model's strengths and limitations in a clinical context.

RQ2. How does the size of the model influence the quality of the answers?

Extended SERVQUAL Evaluation for Clinical Responses

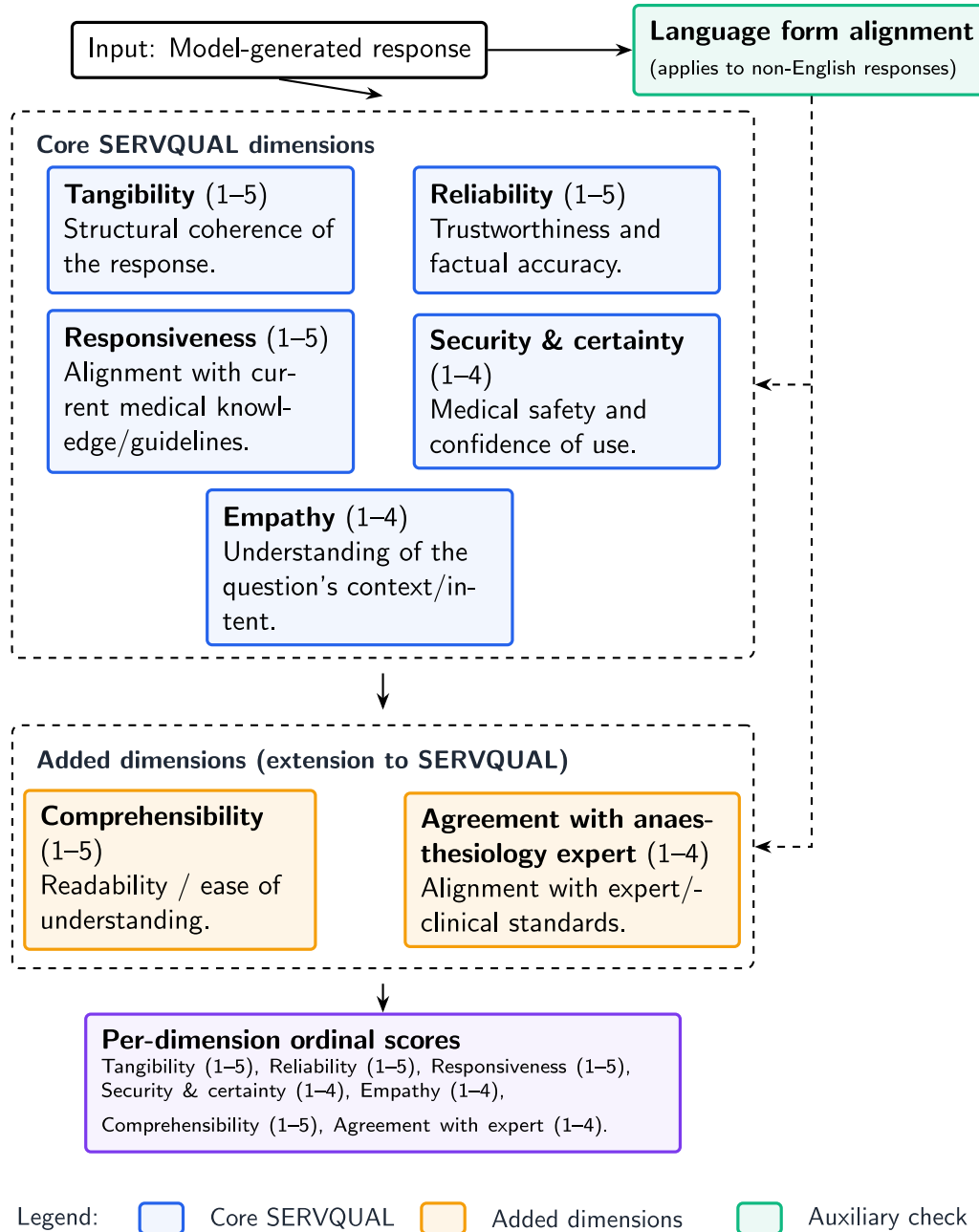


Fig. 2. Diagram of the extended SERVQUAL evaluation framework, showing the five original SERVQUAL dimensions, the two added dimensions, and the auxiliary language-form check applied to non-English responses.

In this research question, we examine the impact of model size on response quality. Using ChatGPT and Gemini as baselines, we compare large models (e.g., 70B parameters), medium-sized models (from 2B to 7B parameters), and small models (up to 2B parameters). The goal is to determine whether smaller models than the commercial ones can produce comparable answers despite limitations in capacity. This comparison will help inform whether more resource-efficient models are viable for specialized medical domains.

RQ3. How accurate are the models across different languages?

This question explores the multilingual capabilities of the models by evaluating their performance on the same set of clinical questions presented in English and Spanish. We assess whether the models can understand and generate appropriate, reliable, and accurate responses

in both languages. This is critical for applications in multilingual clinical environments where access to high-quality information in multiple languages is essential.

RQ4. How efficient are different language models per topic?

Here, we investigate how model performance varies across specific question topics. We categorize the set of questions by subject area and evaluate the performance of each model according to the median of the six core quality metrics introduced in RQ1. This helps identify whether certain models are more effective in specific domains of epidural analgesia, allowing for a more granular understanding of model strengths.

RQ5. What is the intra-LLM variability in answer quality?

Finally, we assess the robustness of each model by analysing the intra-LLM variability —i.e., how much a model's response quality

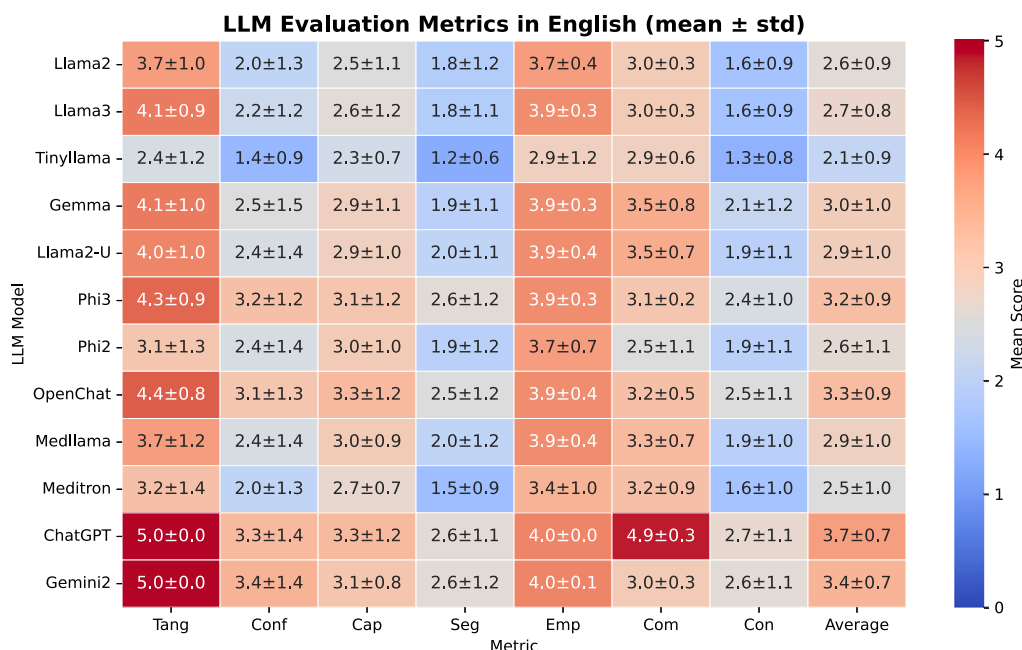


Fig. 3. Mean results of the experimental evaluation for the different LLMs.

fluctuates across different but related questions. We measure this by computing the standard deviation of the evaluation scores obtained in RQ4. A higher standard deviation indicates greater inconsistency, while a lower one suggests more reliable performance across similar inputs.

4.1. Data acquisition and experimental setup

The data was generated using a system designed to send the same question to different large language models. The system uses ten distinct questions and their rephrasing, submitting them to each available LLM. The answers are then evaluated by an anaesthesiologist (the first author), who evaluates them according to the metrics we describe based on an improved version of the SERVQUAL model (see Section 3.3). This evaluation forms the dataset used for our experimental analysis, comprising 100 questions in Spanish and 100 in English, with a total of 2400 answers provided by all LLMs and 16,800 records corresponding to the SERVQUAL metrics. The quantification of the answers' quality follows the numerical mapping described in Section 3.3. The dataset and the code are available on GitHub.²

It is important to note that studies applying the SERVQUAL model in healthcare typically evaluate 22 items [56,57]. The largest item set found in the literature comes from the study that inspired our work, which assesses 240 items [33]. In contrast, this study undertakes a substantially more ambitious evaluation comprising 2400 items.

For the evaluation system, we used a computer running Ubuntu 22.04, equipped with an NVIDIA A30 GPU (24 GB), 128 GB of RAM, and a 200 GB SSD. The system generates all answers in under 24 h. For Gemini and ChatGPT, we accessed the models via their public APIs.

4.2. RQ1 results: Metric values

The first research question focuses on several aspects of answer quality. It begins by measuring *tangibility*, which relates to the structure and construction quality of the answer. It then evaluates *reliability*, aiming to identify which LLMs provide fully trustworthy answers, and when parts of their responses may be unreliable. The third metric assesses the *knowledge* of the LLMs by estimating whether their answers

are knowledgeable about the topic (epidural analgesia) and aligned with the latest guidelines. This helps determine the quality of model training by identifying gaps in knowledge or outdated information.

Next, the fourth metric examines the *safety* and *certainty* of the answers. It evaluates when an LLM could serve as a reliable reference, providing enough certainty for the general public to acquire accurate information—or, conversely, when it might fail to earn public trust. The following metric, *empathy*, assesses whether the model understands the intent behind the questions. Our expert also reviews the *comprehensibility* of the answers, focusing on whether the responses are easy for the general public to understand, difficult to follow, or contain incomprehensible sections. Finally, the last metric measures *concordance with the anaesthesiologist*, checking whether the answers align with expert opinion or diverge partially or significantly. This metric also identifies answers that should never be used for medical decision-making. Fig. 3 and Table 2 present the results of the LLMs across these metrics.

In terms of *tangibility*, the LLMs that obtain the best results are Gemini, ChatGPT, Phi-3, and OpenChat. All of these models produce answers that are well structured and well constructed, achieving values close to the maximum. The models that perform the worst in this regard are TinyLlama, Phi-2, and Meditron. It is important to note that TinyLlama is one of the smallest models, while Phi-2, although medium-sized, is relatively old (released in December 2023). In contrast, Phi-3 (released in April 2024), as the next version, is one of the best performers and demonstrates that a medium-sized model can compete with very large models like Gemini and ChatGPT. Meditron, despite its specialization for medical purposes, achieves poor tangibility scores. While it may not have been trained specifically to generate well-structured answers, its outputs are still readable. It is also noteworthy that Phi-3 and OpenChat, which are much smaller than Gemini and ChatGPT (by about two to three orders of magnitude in terms of parameters), achieve comparable answer quality.

Regarding *reliability or trustworthiness*, no model achieves the highest trust level (5). Most models reach, at best, an approximate normal level (3). The models closest to this level are Gemini, ChatGPT, OpenChat, and Phi-3. The weakest performers include TinyLlama and Meditron, as well as Llama-2, despite being a large model. Given that Llama-2 is unfit in this context, it follows that TinyLlama – trained as a

² <https://github.com/hdg7/epidural>

Table 2

Median results of the experimental evaluation for the different LLMs in English. We applied the Wilcoxon test to assess statistically significant improvements against the commercial models Gemini and ChatGPT, respectively. When the p -value of the test is smaller than 0.01, we reject the null hypothesis and determine whether the results of the current LLM are significantly worse (▼) or better (▲). Otherwise, we cannot reject the null hypothesis (▶).

LLM	Tang	Conf	Cap	Seg	Emp	Com	Con
Llama-2 (GP-L)	▼▼4.0	▼▼1.0	▼▼2.0	▼▼1.0	▼▼4.0	▶▼3.0	▼▼1.0
Llama-3 (GP-L)	▼▼4.0	▼▼2.0	▼▼2.0	▼▼1.0	▼▼4.0	▶▼3.0	▼▼1.0
TinyLlama (GP-S)	▼▼2.0	▼▼1.0	▼▼2.0	▼▼1.0	▼▼4.0	▶▼3.0	▼▼1.0
Gemma (GP-M)	▼▼4.0	▼▼2.0	▶▼3.0	▼▼1.0	▶▼4.0	▶▼3.0	▼▼2.0
Llama-2-Unc. (GP-L)	▼▼4.0	▼▼2.0	▶▼3.0	▼▼1.0	▼▼4.0	▶▼3.0	▼▼1.0
Phi-3 (GP-M)	▼▼5.0	▶▶3.0	▶▶3.0	▶▶2.0	▶▶4.0	▶▼3.0	▶▶2.0
Phi-2 (GP-M)	▼▼3.0	▼▼2.0	▶▶3.0	▼▼1.0	▼▼4.0	▼▼3.0	▼▼1.0
OpenChat (GP-M)	▼▼5.0	▶▶3.0	▶▶3.0	▶▶3.0	▼▼4.0	▶▼3.0	▶▶3.0
MedLlama (SP-L)	▼▼4.0	▼▼2.0	▶▶3.0	▼▼1.0	▼▼4.0	▶▼3.0	▼▼1.0
Meditron (SP-M)	▼▼3.0	▼▼1.0	▼▼3.0	▼▼1.0	▼▼4.0	▶▼3.0	▼▼1.0
ChatGPT (GP-VL)	▶▶5.0	▶▶3.0	▶▶3.0	▶▶3.0	▶▶4.0	▶▶5.0	▶▶3.0
Gemini (GP-VL)	▶▶5.0	▶▶4.0	▶▶3.0	▶▶3.0	▶▶4.0	▶▼3.0	▶▶3.0

reduced version of Llama-2 – is also inadequate. Since this metric evaluates reliability, most cases reflect a normal level of trustworthiness, but none of the answers are fully precise and completely reliable.

With respect to the capacity to provide **updated knowledge**, none of the large language models approach the ideal level (5). All achieve, at most, normal levels (3). This suggests that while the LLMs provide reasonable knowledge, they do not fully align with the latest guidelines or contain the most recent information, which could be valuable for users seeking accurate medical guidance. Models like Llama-3, Llama-2, and TinyLlama particularly lack up-to-date knowledge, scoring the lowest (2). The shared lineage of these models demonstrates how their related architecture and training impact their performance in this evaluation.

Regarding the **security and certainty** of the answers, Gemini, ChatGPT, and OpenChat are the most secure (reaching up to 3). Although none of these systems provide answers that can be considered a full reference (4), they typically produce responses that are sufficiently certain and reliable for the general public to trust (3). Notably, OpenChat performs on par with the very large commercial LLMs while requiring significantly fewer computational resources. Similarly, Phi-3 reaches normal levels of quality (2), making it a viable option as an informative resource. The remaining models provide answers that are neither trustworthy nor suitable for the general public when seeking information about epidural anaesthesia.

In terms of **empathy**, all LLMs produce answers that demonstrate an understanding of the question's meaning, confirming their quality as language models. However, there are some reservations about models like TinyLlama and Meditron, whose responses occasionally lack sufficient sensitivity or contextual awareness.

With respect to **comprehension**, the best model is ChatGPT (5), clearly standing out as its answers are very easy for the general public to understand. The remaining models generally provide responses with a normal level of comprehensibility (3).

Finally, in terms of **concordance with the expert anaesthesiologist**, ChatGPT, Gemini, and OpenChat are again the top performers (3), although none provide perfect concordance. Gemma and Phi-3 show statistics indicating approximately 50% concordance, while the rest are generally below this threshold. Overall, most models align with the expert's response in about 50% of their answers. The stronger models exceed this threshold (3), while the average ones achieve lower results (2). It is important to emphasize that models at the lower level (2) should never be used for medical decision-making or treatment guidance.

RQ1 Answer. The models' quality vary depending on the evaluation metric. However, the best model according to all quality metrics is ChatGPT, followed by Gemini. Among the openly available models, the best in the context of epidural anaesthesia are Phi-3 and OpenChat.

4.3. RQ2 results: Size and quality of the answers

The second research question asks about the influence of model size on the quality of large language models. We present comparative results in Fig. 3 and Table 2. From these, we observe that the **very large models** (ChatGPT and Gemini) do not show substantial differences overall. However, ChatGPT's comprehension notably exceeds that of Gemini (with a median comprehension of 5 versus 3), making it one of the top models for answering questions about epidural anaesthetics.

For the **large general-purpose models**, Llama-2, Llama-3, and Llama-2-Unc. achieve similar results overall. However, Llama-2 performs worse in terms of reliability (median of 1 versus a median of 2 for the others), while Llama-2-Unc. performs better in responsiveness (median of 3 versus 2 for the others), possibly due to its training with more medical knowledge. Llama-3 and Llama-2 also outperform Gemini in comprehensibility. Compared to all models, these general-purpose models are statistically inferior to ChatGPT in almost every metric except responsiveness, where Llama-2-Unc. matches ChatGPT (2.9 vs 3.3, with no statistically significant difference according to the Wilcoxon test). Compared to Gemini, they are generally worse in most metrics, except comprehensibility, where they obtain similar results (median of 3, no statistically significant difference). Notably, Llama-2-Unc. even outperforms Gemini in comprehensibility (3.5 vs 3.0, with a statistically significant improvement) and matches it in reliability (2.9 vs 3.1, no significant difference). The large specialized model MedLlama performs similarly to Llama-2-Unc..

For the **medium-sized models**, Gemma achieves better or comparable results to Llama-2-Unc., despite using fewer resources. Its responsiveness is better than Gemini's (3.5 vs 3.0, with statistically significant improvement), and its comprehensibility also exceeds that of Llama-2-Unc. (2.1 vs 1.9). Phi-2 performs similarly but slightly worse than Llama-2—worse in tangibility (3.1 vs 3.7) and comprehensibility (2.5 vs 3.0), but better in confidentiality (2.4 vs 2.0) and responsiveness (2.5 vs 3), comparable to ChatGPT and Gemini (with no statistically significant difference). Phi-3, as an improved version of Phi-2, achieves significantly better results, especially in tangibility (4.3 vs 3.1), as well as across all other metrics. It performs comparably to Gemini, with no significant differences except in tangibility, where Gemini is slightly better (5.0 vs 4.3). This demonstrates that a medium-sized general-purpose model, suitable for systems with limited GPU resources, can

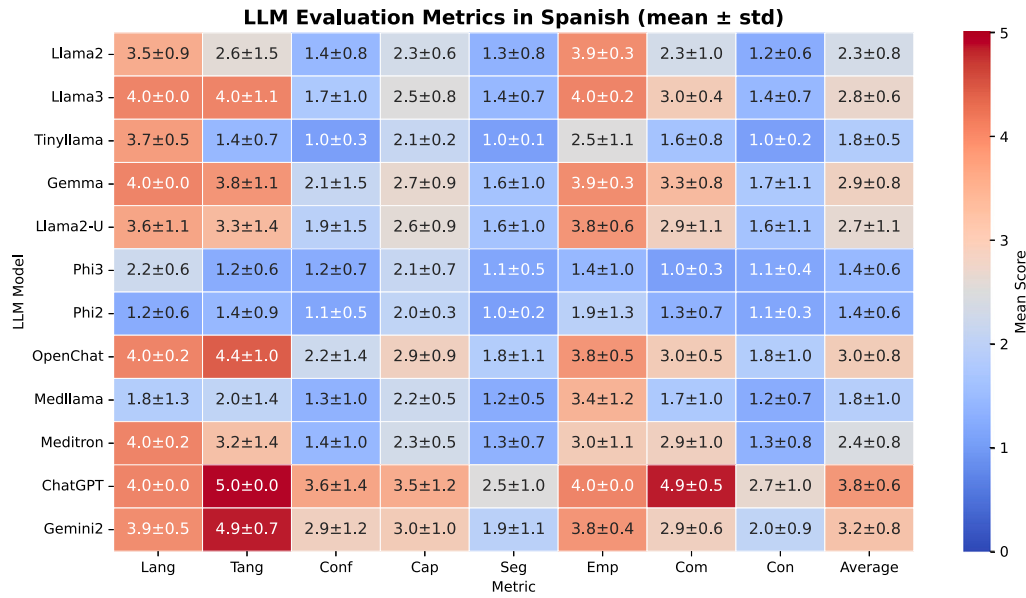


Fig. 4. Mean and standard deviation results of the experimental evaluation for the different LLMs in Spanish.

Table 3

Median results of the experimental evaluation for the different LLMs in Spanish. We applied the Wilcoxon test to assess statistically significant improvements against the commercial models Gemini and ChatGPT, respectively. When the p -value of the test is smaller than 0.01, we reject the null hypothesis and determine whether the results of the current LLM are significantly worse ($\blacktriangledown$) or better ($\blacktriangle$). Otherwise, we cannot reject the null hypothesis ($\blacktriangleright$).

LLM	Lang	Tang	Conf	Cap	Seg	Emp	Com	Con
Llama-2 (GP-L)	$\blacktriangledown$ 4.0	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangleright$ 4.0	$\blacktriangledown$ 3.0	$\blacktriangledown$ 1.0
Llama-3 (GP-L)	$\blacktriangleright$ 4.0	$\blacktriangledown$ 4.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangleright$ 4.0	$\blacktriangleright$ 3.0	$\blacktriangledown$ 1.0
TinyLlama (GP-S)	$\blacktriangledown$ 4.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 3.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0
Gemma (GP-M)	$\blacktriangleright$ 4.0	$\blacktriangledown$ 4.0	$\blacktriangledown$ 1.0	$\blacktriangleright$ 2.0	$\blacktriangleright$ 1.0	$\blacktriangleright$ 4.0	$\blacktriangleright$ 3.0	$\blacktriangleright$ 1.0
Llama-2-Unc. (GP-L)	$\blacktriangledown$ 4.0	$\blacktriangledown$ 3.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 2.0	$\blacktriangleright$ 1.0	$\blacktriangleright$ 4.0	$\blacktriangleright$ 3.0	$\blacktriangledown$ 1.0
Phi-3 (GP-M)	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0
Phi-2 (GP-M)	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0
OpenChat (GP-M)	$\blacktriangleright$ 4.0	$\blacktriangledown$ 5.0	$\blacktriangledown$ 2.0	$\blacktriangleright$ 3.0	$\blacktriangleright$ 1.0	$\blacktriangleright$ 4.0	$\blacktriangleright$ 3.0	$\blacktriangleright$ 1.0
MedLlama (SP-L)	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangleright$ 4.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 1.0
Meditron (SP-M)	$\blacktriangleright$ 4.0	$\blacktriangledown$ 3.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 2.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 3.0	$\blacktriangleright$ 3.0	$\blacktriangledown$ 1.0
ChatGPT (GP-VL)	$\blacktriangleright$ 4.0	$\blacktriangleright$ 5.0	$\blacktriangleright$ 4.0	$\blacktriangleright$ 4.0	$\blacktriangleright$ 3.0	$\blacktriangleright$ 4.0	$\blacktriangleright$ 5.0	$\blacktriangleright$ 3.0
Gemini (GP-VL)	$\blacktriangleright$ 4.0	$\blacktriangleright$ 5.0	$\blacktriangleright$ 3.0	$\blacktriangleright$ 3.0	$\blacktriangledown$ 1.0	$\blacktriangledown$ 4.0	$\blacktriangleright$ 3.0	$\blacktriangledown$ 2.0

match the performance of one of the largest and most competitive models on the market, while outperforming larger models that consume more computational resources. OpenChat also delivers strong results, similar to Phi-3, and slightly better in concordance with anaesthetic experts (2.4 vs 2.5 mean; 2 vs 3 median), making it a strong candidate for use in clinical decision or advice support systems within this domain. Conversely, the model source still matters, as illustrated by Meditron, whose results are worse than those of Phi-3 and OpenChat, despite being trained with medical data. It also underperforms compared to its larger counterpart, MedLlama.

Finally, **the smallest model**, TinyLlama, performs the worst across all metrics. Although this could be attributed to its limited capacity, its performance may also reflect its training origin, as it is a reduced version of Llama-2. Indeed, Llama-2 performs similarly to TinyLlama in median for most metrics except tangibility (4 vs 2).

RQ2 Answer. Although ChatGPT remains the best and most reliable model in terms of the size–performance trade-off, two medium models –OpenChat and Phi-3– achieve the best comparative results, significantly outperforming any larger model. This highlights that training quality outweighs model size.

Regarding how different languages affect model capabilities, multilingual support remains a significant limitation of large language

models. This is evident in every case except for ChatGPT, where results worsen for all other models (see Fig. 4 and Table 3).

First, in terms of language synthesis, many models do not provide their answers in the queried language. This is reflected in the column “Lang” in Fig. 4 and Table 3. This value measures how often the answer is in the same language as the question (4), in the same language but with mistakes (3), a mix of languages (2), or in English (1). The models that struggle most to maintain the language during their answers are MedLlama (1, typically answering in English), Phi-2 (2, usually mixing languages), and Phi-3 (1). In contrast, Llama-2 (3.5), TinyLlama (3.7), and Llama-2-Unc. (3.6) show lower performance in Spanish, but generally answer in the correct language. This highlights the importance of ensuring that a model can operate effectively in the appropriate linguistic context. For example, Phi-3 might be a good choice for informing the general public about epidural treatment (see Section 4.3), but its reliability is limited by the language used to query it. One of the most interesting cases is ChatGPT, which improves the quality of its responses when using Spanish instead of English. Answers in Spanish retain the strengths of their English equivalents while becoming more reliable (with median reliability improving from 3 to 4) and better aligned with current knowledge (responsiveness increasing from 3 to 4).

In contrast, Phi-2 and Phi-3 become almost completely incoherent when the language changes, with nearly all metrics dropping to a

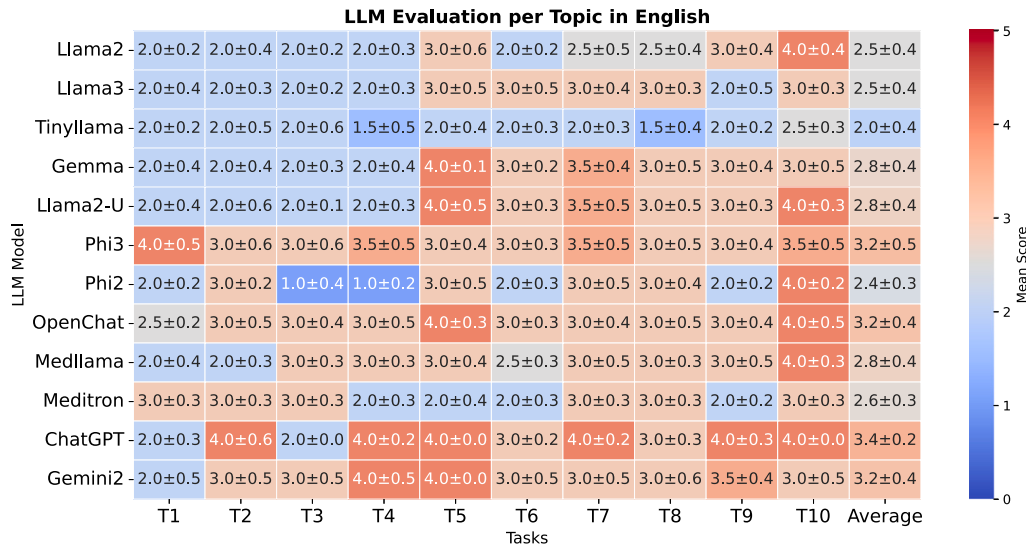


Fig. 5. Median and standard deviation of the average quality for the different answer in English aggregated per topic.

median of 1, making them among the worst-performing models. OpenChat achieves the best statistics among all open models (large, medium, and small), but its capabilities decline in terms of reliability (from 3 to 2), safety (from 3 to 1), and concordance with experts (from 3 to 1). These results make OpenChat unsuitable for providing information about epidural anaesthetics to the general public. Similarly, Gemini also loses capability across these metrics, making it less suitable for this task.

RQ3 Answer. The capabilities of large language models in multilingual scenarios (concretely Spanish and English) are significantly affected by the language, making most of them unsuitable for the task regardless of their size—from small models to very large ones such as Gemini. The only model that not only remains suitable but actually improves in this context is ChatGPT, which becomes more up-to-date and reliable in Spanish. OpenChat achieves the best results among open models, but its capabilities do not make it a suitable choice for providing answers about epidural anaesthetics in Spanish. It is also important to note that specialized models do not perform better in the multilingual setting.

4.4. RQ4 results: Results per topic

Research Question 4 focuses on the specific topics of the different questions. Figs. 5 and 6 show the mean results per topic in each language. In general, ChatGPT has the best performance across all questions, though its performance typically decreases on harder questions. When we analyse each of the different topics, we can see that certain types of questions challenge the models differently depending on the language model. The language of the question also affects model performance.

For the **first topic**, analysing the questions and answers in English reveals varied results across models. This first question is controversial. For example, Phi-3 obtained the best result (4), but its performance declines in Spanish. Phi-3 is followed by Meditron (3) and OpenChat (2.5). ChatGPT's quality is lower (2) on this specific topic, making it less reliable. In Spanish, Phi-3's performance drops slightly (3). In this case, Llama-2-Unc. and OpenChat achieved the best results (4). ChatGPT also improves on this topic (from 2 to 3). Other models, such as Gemma (3.8) and Gemini (3.2), also outperformed ChatGPT.

For the **second topic**, which requires a simple and direct answer, ChatGPT is the best model (4). The second-best models are Phi-3,

Phi-2, OpenChat, MedLlama, Meditron, and Gemini (3). In Spanish, these results change: ChatGPT remains the best model, but Phi-2 and Phi-3 models perform poorly (no higher than 1), and most models show a performance drop—except for Gemini (3), which remains the second-best.

The **third topic** is designed to be confusing, as it asks about long-term epidural problems. The models tend to be misled by the long-term aspect, and none achieve 4 points. There is a large tie at 3 points between Phi-3, OpenChat, MedLlama, Meditron, and Gemini. In this case, ChatGPT's performance drops (2). However, when we analyse the results in Spanish, ChatGPT improves on this specific question (3.5), while all other models decline. The Phi-2 and Phi-3 models continue to drop in performance, as does MedLlama (1). Other models also drop slightly; for example, OpenChat and Meditron drop to 2.5, while Gemini remains at 3.

For the **fourth topic**, which also requires a simple and direct answer, ChatGPT and Gemini obtain the best results (4), followed by Phi-3 (3.5), OpenChat (3), and MedLlama (3). In Spanish, ChatGPT excels (4.5), while Gemini performs slightly worse (3). OpenChat and Gemma tie (3), and the other models decline, as in the previous cases.

The **fifth topic's answer** is also simple and direct. It is the easiest topic for many of the models and asks who administers the epidural during childbirth. Gemma, Llama-2-Unc., OpenChat, Gemini, and ChatGPT all obtain good results (4). The Phi-2 and Phi-3 models, MedLlama, Llama-2, and Llama-3 achieve slightly lower but reasonable results (3). In Spanish, Gemma, Llama-2-Unc., OpenChat, ChatGPT, and Gemini again obtain good results (4), while MedLlama, Llama-2, and Llama-3 maintain their quality (3), and the Phi-2 and Phi-3 models drop to 1, as seen previously.

The **sixth topic** is difficult. It concerns what happens if a person moves during the epidural injection. To provide an accurate answer, the model needs to incorporate a broad range of knowledge, which makes the task challenging. The models follow a pattern similar to Topic 3. None of the models achieve 4 points, with the maximum at 3, where there is a general tie among Llama-3, Gemma, Llama-2-Unc., Phi-3, OpenChat, ChatGPT, and Gemini. In Spanish, ChatGPT performs best (3.5), Gemini maintains its quality (3), and OpenChat remains at a similar level. The rest drop, following patterns observed in earlier topics.

For the **seventh topic**, where the answer is simple and direct, ChatGPT performs best (4), followed by Gemma, Llama-2, and Phi-3 (3.5). The other models obtain good results around 3. In Spanish, ChatGPT is again the best (4). Gemma, Llama-2, Llama-2-Unc., and OpenChat drop slightly (3), while Gemini maintains its result (3).

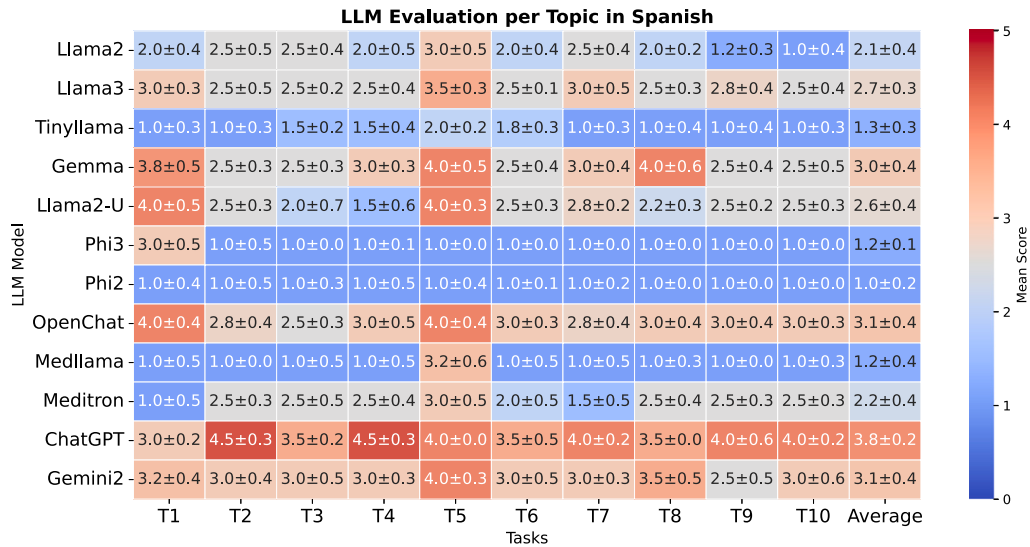


Fig. 6. Median and standard deviation of the average quality for the different answer in Spanish aggregated per topic.

For **Topic 8**, which is slightly harder, the models follow a pattern similar to Topic 6. There are no clear standouts, with most models tying at around 3. In Spanish, Gemma performs best (4), followed by ChatGPT and Gemini (3.5), while the others show a decline in performance.

For **Topic 9**, which has a simple and direct answer, ChatGPT performs best in English (4), followed by Gemini (3.5). The other models achieve results between 2 and 3. In Spanish, ChatGPT remains the best (4). All other models drop below 3, except OpenChat, which maintains its quality level (3).

Finally, the **last topic** has a simple and direct answer. In English, several models achieve good results, including ChatGPT, Llama-2, Llama-2-Unc., Phi-2, OpenChat, and Llama-3, with scores close to 3 or 3.5. In Spanish, all models drop except ChatGPT (4) and Gemini (3).

RQ4 Answer. The difficulty of the topics affects the quality of the responses, as the most complex or controversial topics produce the poorest results. This makes the models less reliable when answering sensitive questions. The best-performing models across different topics are ChatGPT, Gemini, OpenChat, and Phi-3 in English, and ChatGPT and Gemini in Spanish. OpenChat remains reliable for certain topics in Spanish, although its results decline overall.

4.5. RQ5 results: Intra-LLM variability

Research Question 5 focuses on internal quality stability per question. Figs. 5 and 6 present these results by calculating the standard deviation of answer quality by topic. Topics with a higher standard deviation exhibit answers that are more diverse in quality, while topics with a lower standard deviation have answers that are more stable in terms of quality. ChatGPT and Phi-2 tend to be the most stable models for almost every topic in English, with standard deviations lower than or equal to 0.2 in the majority of cases. However, while ChatGPT achieves high answer quality, Phi-2's quality is lower according to the generated answers. The other models vary considerably, showing significant quality fluctuations depending on how the questions are phrased.

The topics with the most stable answers are Topics 1, 3, 5, 6, and 9. Unfortunately, these do not follow any clear pattern regarding question difficulty or whether the topics are controversial. The least stable topics

are 2, 7, and 8, with Topic 8 being the least stable overall. Although in Spanish the answers tend to be more stable, this appears to result from declining quality, particularly in the cases of Phi-3 and Phi-2. This trend does not apply to ChatGPT, which improves its quality while maintaining stability. In this case as well, no patterns emerge related to the difficulty of the questions.

RQ5 Answer. The intra-LLM variability is generally greater than 0.2 in most models, except for ChatGPT, which shows the lowest variation, making it the most stable model in both languages. In terms of topics, no clear patterns emerge to identify when models are more stable.

4.6. Confidence intervals, effect size, and FDR adjustment

We have incorporated additional statistical analyses to strengthen the robustness of our results. First, we calculated the effect size for all comparisons performed using the Wilcoxon test. This was done following the methodology proposed by Fritz, Morris, and Richler [58] and extended by Tomczak and Tomczak [59]. According to these authors, the effect size r is computed as $r = Z/\sqrt{n}$, where Z is the Wilcoxon test's Z-score and n is the total number of observations. Consistent with the literature, we classified effect sizes into three categories: small ($r < 0.3$), medium ($0.3 \leq r < 0.5$), and large ($r \geq 0.5$).

For the 77 English comparisons with Gemini, 74 showed a large effect size and 3 a medium effect. The results for ChatGPT were identical: 74 large effects and 3 medium effects. For the 88 Spanish comparisons with both ChatGPT and Gemini, all comparisons exhibited a large effect size.

To ensure proper adjustment for multiple comparisons, we applied a False Discovery Rate (FDR) correction using the Benjamini–Hochberg (BH) procedure to the p-values. After adjustment, only one p-value changed substantially: the English comparison between Gemma and ChatGPT on empathy, which became non-significant. This change is reflected in Table 2.

We also provide the confidence intervals for both the general results in English and Spanish (Tables 4 and 5) and for the comparison in both languages with the best model, i.e., ChatGPT (Tables 6 and 7). Overall, the intervals confirm the dominance of ChatGPT and Gemini2 in most dimensions, with both models reaching upper bounds of 5.0 in Tangibility and Communication in English and Spanish. In contrast, smaller models such as TinyLlama and Phi-2 and Phi-3 variants

Table 4
95% confidence intervals in English.

LLM	Tang	Conf	Cap	Seg	Emp	Com	Con
Llama-2	[3.5, 3.9]	[1.7, 2.3]	[2.3, 2.7]	[1.6, 2.0]	[3.6, 3.8]	[2.9, 3.1]	[1.4, 1.8]
Llama-3	[3.9, 4.3]	[2.0, 2.4]	[2.4, 2.8]	[1.6, 2.0]	[3.8, 4.0]	[2.9, 3.1]	[1.4, 1.8]
TinyLlama	[2.2, 2.6]	[1.2, 1.6]	[2.2, 2.4]	[1.1, 1.3]	[2.7, 3.1]	[2.8, 3.0]	[1.1, 1.5]
Gemma	[3.9, 4.3]	[2.2, 2.8]	[2.7, 3.1]	[1.7, 2.1]	[3.8, 4.0]	[3.3, 3.7]	[1.9, 2.3]
Llama-2-Unc.	[3.8, 4.2]	[2.1, 2.7]	[2.7, 3.1]	[1.8, 2.2]	[3.8, 4.0]	[3.4, 3.6]	[1.7, 2.1]
Phi-3	[4.1, 4.5]	[3.0, 3.4]	[2.9, 3.3]	[2.4, 2.8]	[3.8, 4.0]	[3.1, 3.1]	[2.2, 2.6]
Phi-2	[2.8, 3.4]	[2.1, 2.7]	[2.8, 3.2]	[1.7, 2.1]	[3.6, 3.8]	[2.3, 2.7]	[1.7, 2.1]
OpenChat	[4.2, 4.6]	[2.8, 3.4]	[3.1, 3.5]	[2.3, 2.7]	[3.8, 4.0]	[3.1, 3.3]	[2.3, 2.7]
MedLlama	[3.5, 3.9]	[2.1, 2.7]	[2.8, 3.2]	[1.8, 2.2]	[3.8, 4.0]	[3.2, 3.4]	[1.7, 2.1]
Meditron	[2.9, 3.5]	[1.7, 2.3]	[2.6, 2.8]	[1.3, 1.7]	[3.2, 3.6]	[3.0, 3.4]	[1.4, 1.8]
ChatGPT	[5.0, 5.0]	[3.0, 3.6]	[3.1, 3.5]	[2.4, 2.8]	[4.0, 4.0]	[4.8, 5.0]	[2.5, 2.9]
Gemini	[5.0, 5.0]	[3.1, 3.7]	[2.9, 3.3]	[2.4, 2.8]	[4.0, 4.0]	[2.9, 3.1]	[2.4, 2.8]

Table 5
95% confidence intervals in Spanish.

LLM	Lang	Tang	Conf	Cap	Seg	Emp	Com	Con
Llama-2	[3.3, 3.7]	[2.3, 2.9]	[1.2, 1.6]	[2.2, 2.4]	[1.1, 1.5]	[3.8, 4.0]	[2.1, 2.5]	[1.1, 1.3]
Llama-3	[4.0, 4.0]	[3.8, 4.2]	[1.5, 1.9]	[2.3, 2.7]	[1.3, 1.5]	[4.0, 4.0]	[2.9, 3.1]	[1.3, 1.5]
TinyLlama	[3.6, 3.8]	[1.3, 1.5]	[0.9, 1.1]	[2.1, 2.1]	[1.0, 1.0]	[2.3, 2.7]	[1.4, 1.8]	[1.0, 1.0]
Gemma	[4.0, 4.0]	[3.6, 4.0]	[1.8, 2.4]	[2.5, 2.9]	[1.4, 1.8]	[3.8, 4.0]	[3.1, 3.5]	[1.5, 1.9]
Llama-2-Unc.	[3.4, 3.8]	[3.0, 3.6]	[1.6, 2.2]	[2.4, 2.8]	[1.4, 1.8]	[3.7, 3.9]	[2.7, 3.1]	[1.4, 1.8]
Phi-3	[2.1, 2.3]	[1.1, 1.3]	[1.1, 1.3]	[2.0, 2.2]	[1.0, 1.2]	[1.2, 1.6]	[0.9, 1.1]	[1.0, 1.2]
Phi-2	[1.1, 1.3]	[1.2, 1.6]	[1.0, 1.2]	[1.9, 2.1]	[1.0, 1.0]	[1.6, 2.2]	[1.2, 1.4]	[1.0, 1.2]
OpenChat	[4.0, 4.0]	[4.2, 4.6]	[1.9, 2.5]	[2.7, 3.1]	[1.6, 2.0]	[3.7, 3.9]	[2.9, 3.1]	[1.6, 2.0]
MedLlama	[1.5, 2.1]	[1.7, 2.3]	[1.1, 1.5]	[2.1, 2.3]	[1.1, 1.3]	[3.2, 3.6]	[1.5, 1.9]	[1.1, 1.3]
Meditron	[4.0, 4.0]	[2.9, 3.5]	[1.2, 1.6]	[2.2, 2.4]	[1.2, 1.4]	[2.8, 3.2]	[2.7, 3.1]	[1.1, 1.5]
ChatGPT	[4.0, 4.0]	[5.0, 5.0]	[3.3, 3.9]	[3.3, 3.7]	[2.3, 2.7]	[4.0, 4.0]	[4.8, 5.0]	[2.5, 2.9]
Gemini	[3.8, 4.0]	[4.8, 5.0]	[2.7, 3.1]	[2.8, 3.2]	[1.7, 2.1]	[3.7, 3.9]	[2.8, 3.0]	[1.8, 2.2]

Table 6
95% confidence intervals for mean differences (Model vs ChatGPT) in English.

LLM	Tang	Conf	Cap	Seg	Emp	Com	Con
Llama-2	[-1.5, -1.1]	[-1.7, -0.9]	[-1.1, -0.5]	[-1.1, -0.5]	[-0.4, -0.2]	[-2.0, -1.8]	[-1.4, -0.8]
Llama-3	[-1.1, -0.7]	[-1.5, -0.7]	[-1.0, -0.4]	[-1.1, -0.5]	[-0.2, -0.0]	[-2.0, -1.8]	[-1.4, -0.8]
TinyLlama	[-2.8, -2.4]	[-2.2, -1.6]	[-1.3, -0.7]	[-1.6, -1.2]	[-1.3, -0.9]	[-2.1, -1.9]	[-1.7, -1.1]
Gemma	[-1.1, -0.7]	[-1.2, -0.4]	[-0.7, -0.1]	[-1.0, -0.4]	[-0.2, -0.0]	[-1.6, -1.2]	[-0.9, -0.3]
Llama-2-Unc.	[-1.2, -0.8]	[-1.3, -0.5]	[-0.7, -0.1]	[-0.9, -0.3]	[-0.2, -0.0]	[-1.6, -1.2]	[-1.1, -0.5]
Phi-3	[-0.9, -0.5]	[-0.5, 0.3]	[-0.5, 0.1]	[-0.3, 0.3]	[-0.2, -0.0]	[-1.9, -1.7]	[-0.6, -0.0]
Phi-2	[-2.2, -1.6]	[-1.3, -0.5]	[-0.6, 0.0]	[-1.0, -0.4]	[-0.4, -0.2]	[-2.6, -2.2]	[-1.1, -0.5]
OpenChat	[-0.8, -0.4]	[-0.6, 0.2]	[-0.3, 0.3]	[-0.4, 0.2]	[-0.2, -0.0]	[-1.8, -1.6]	[-0.5, 0.1]
MedLlama	[-1.5, -1.1]	[-1.3, -0.5]	[-0.6, -0.0]	[-0.9, -0.3]	[-0.2, -0.0]	[-1.8, -1.4]	[-1.1, -0.5]
Meditron	[-2.1, -1.5]	[-1.7, -0.9]	[-0.9, -0.3]	[-1.4, -0.8]	[-0.8, -0.4]	[-1.9, -1.5]	[-1.4, -0.8]
Gemini	[0.0, 0.0]	[-0.3, 0.5]	[-0.5, 0.1]	[-0.3, 0.3]	[-0.0, 0.0]	[-2.0, -1.8]	[-0.4, 0.2]

show consistently narrow and low ranges, especially in Confidence and Segmentation, where intervals remain below 1.6 in Spanish. When comparing against ChatGPT, the differences are striking: most models exhibit negative intervals across all metrics, indicating significantly lower performance. For example, TinyLlama shows the largest gaps in Tangibility ([-2.8, -2.4] in English and [-3.7, -3.5] in Spanish) and Communication ([-2.1, -1.9] and [-3.5, -3.1], respectively). Interestingly, Gemini2 is the only model approaching parity, with intervals near zero for Tangibility and Language in Spanish, suggesting competitive performance in these areas. These results reinforce that while several models perform well in empathy and comprehension, none surpass ChatGPT, and only Gemini2 comes close in specific tasks.

4.7. Discussion: Trustworthiness of the answers

We evaluate the trustworthiness of the answers, focusing on whether they are capable of influencing or changing a person's opinion. Given the inherently subjective nature of this assessment, we use the SERVQUAL model to define rules that allow us to automate a pre-screening of this scenario. We consider three categories of answers:

1. **Invalid answers**, which are not properly constructed or do not meet the minimum criteria for quality.

2. **Trusted answers**, which are not only well-formed but also reliable and suitable for a patient to trust.
3. **Potentially mind-changing answers**, which appear trustworthy on the surface but are unreliable in terms of the information they provide. These answers may lead patients to form incorrect beliefs or change their understanding based on misleading content.

Concretely, we consider an answer invalid when: *tangibility* is below normal (3), *empathy* is below normal (2), the language is not correct (smaller than 4) or the answer does not respond to the question.

We only classify an answer as trusted or not trusted when it is sufficiently *valid* to be understood and when the system clearly addresses the question. An answer is considered trusted when it meets all three conditions:

Agreement with the anaesthesiology expert is 50% or above (3), *security* is normal or higher (3), *reliability* is normal or higher (3), and *responsiveness* is at least normal (3).

If an answer is well-written and valid but fails to meet these previous criteria, we classify it as non-trusted. Such answers could potentially influence a patient's perception in misleading ways.

We can observe in Table 8 that all models are less trustworthy in Spanish than in English. However, ChatGPT slightly increases the

Table 7

95% confidence intervals for mean differences (Model vs ChatGPT) in Spanish.

LLM	Lang	Tang	Conf	Cap	Seg	Emp	Com	Con
Llama-2	[-0.7, -0.3]	[-2.7, -2.1]	[-2.5, -1.9]	[-1.5, -0.9]	[-1.5, -1.0]	[-0.2, 0.0]	[-2.8, -2.4]	[-1.7, -1.3]
Llama-3	[0.0, 0.0]	[-1.2, -0.8]	[-2.2, -1.6]	[-1.3, -0.7]	[-1.3, -0.9]	[-0.0, 0.0]	[-2.0, -1.8]	[-1.5, -1.1]
TinyLlama	[-0.4, -0.2]	[-3.7, -3.5]	[-2.9, -2.3]	[-1.6, -1.2]	[-1.7, -1.3]	[-1.7, -1.3]	[-3.5, -3.1]	[-1.9, -1.5]
Gemma	[0.0, 0.0]	[-1.4, -1.0]	[-1.9, -1.1]	[-1.1, -0.5]	[-1.2, -0.6]	[-0.2, 0.0]	[-1.8, -1.4]	[-1.3, -0.7]
Llama-2-Unc.	[-0.6, -0.2]	[-2.0, -1.4]	[-2.1, -1.3]	[-1.2, -0.6]	[-1.2, -0.6]	[-0.3, -0.1]	[-2.2, -1.8]	[-1.4, -0.8]
Phi-3	[-1.9, -1.7]	[-3.9, -3.7]	[-2.7, -2.1]	[-1.7, -1.1]	[-1.6, -1.2]	[-2.8, -2.4]	[-4.0, -3.8]	[-1.8, -1.4]
Phi-2	[-2.9, -2.7]	[-3.8, -3.4]	[-2.8, -2.2]	[-1.7, -1.3]	[-1.7, -1.3]	[-2.4, -1.8]	[-3.8, -3.4]	[-1.8, -1.4]
OpenChat	[-0.0, 0.0]	[-0.8, -0.4]	[-1.8, -1.0]	[-0.9, -0.3]	[-1.0, -0.4]	[-0.3, -0.1]	[-2.0, -1.8]	[-1.2, -0.6]
MedLlama	[-2.5, -1.9]	[-3.3, -2.7]	[-2.6, -2.0]	[-1.6, -1.0]	[-1.5, -1.1]	[-0.8, -0.4]	[-3.4, -3.0]	[-1.7, -1.3]
Meditron	[-0.0, 0.0]	[-2.1, -1.5]	[-2.5, -1.9]	[-1.5, -0.9]	[-1.4, -1.0]	[-1.2, -0.8]	[-2.2, -1.8]	[-1.6, -1.2]
Gemini	[-0.2, -0.0]	[-0.2, 0.0]	[-1.1, -0.3]	[-0.8, -0.2]	[-0.9, -0.3]	[-0.3, -0.1]	[-2.1, -1.9]	[-1.0, -0.4]

Table 8

Model outcomes by trust category.

Model	Trusted Eng	Untrusted Eng	Invalid Eng	Trusted Spa	Untrusted Spa	Invalid Spa
ChatGPT	52	47	1	56	44	0
OpenChat	46	52	2	22	72	6
Gemini	44	45	11	25	68	7
Phi-3	44	53	3	2	3	95
Gemma	33	60	7	22	65	13
Llama-2-Unc.	27	66	7	20	55	25
MedLlama	25	52	23	1	17	82
Llama-2	20	68	12	2	39	59
Llama-3	20	75	5	9	81	10
Phi-2	18	36	46	0	1	99
Meditron	11	43	46	6	49	45
TinyLlama	7	28	64	0	7	93

number of trustworthy answers when switching to Spanish. Another notable trend is the significant rise in invalid answers across most models when moving from English to Spanish. There are cases, such as Phi-2, in which almost every single answer is invalid, likely because the model does not generate output in the language requested by the user. A similar pattern appears in Phi-3, MedLlama, and TinyLlama. Regarding untrusted answers, some results are particularly interesting. Models such as OpenChat, Gemini, Gemma, Llama-3, and Meditron increase their number of untrusted answers in Spanish. In these cases, the systems not only respond in the correct language but also provide answers that are potentially misleading users or influencing their understanding of the information presented.

It is important to recall that these results are derived following the rules of the SERVQUAL methodology. Of course, trustworthiness is inherently subjective and strongly influenced by a patient's cultural background, meaning that a future human study will be required to validate these interpretations. We acknowledge this as a limitation of the present analysis. Nonetheless, the SERVQUAL system provides a valuable pre-screening tool for assessing how potentially dangerous different models can be. Combined with appropriate human studies, it can offer an approximation of whether a system should be used in the context of epidural anaesthesia or other medical conditions. It is essential to emphasize that this type of evaluation must be performed independently for each condition; results from epidural analgesia cannot be generalized to other treatments, such as cancer therapies. Different models may be more suitable for some medical contexts than others, and for now, each scenario must be evaluated separately to understand these differences.

4.8. Discussion: Clinical implications

Although our results indicate that medium-sized models can outperform some larger models, obstetricians and anaesthesiologists should interpret these findings with caution before considering their application in practice. We outline the key considerations as follows. Medium-sized models may be deployed as decision-support tools, particularly in resource-limited settings, provided that their outputs are reviewed

and validated by human experts. This approach could improve access to reliable information while maintaining clinical safety and oversight. However, prior evaluation within each relevant domain of expertise is essential to validate the system before deployment, and identify common errors. The most common errors can be classified according to recurring issues such as incoherences, mistranslations, and factual inaccuracies. This section presents some of the most problematic and potentially harmful responses within the category of factual inaccuracies, as produced by the evaluated models. These examples highlight critical concerns regarding the reliability of LLM-generated medical advice for the general public. When prompted in English, some of the most concerning responses included:

- **Meditron:** "It is not possible to get an epidural when giving birth. It is only possible at the end of labour, if it is necessary". When asked when it is safe to request an epidural during childbirth, Meditron responded: "In the last 2 weeks, before the contractions begin". It also incorrectly listed long-term risks of epidural use as including pneumothorax, increased risk of infection, and wound dehiscence.
- **MedLlama:** In response to the question "What are the most frequent complications women experience with epidurals?", it stated that the most common complication is a drop in blood pressure, which may lead to fainting or even a heart attack.
- **TinyLlama:** When asked whether there is a specific time window during childbirth to request an epidural, it stated that an epidural can be requested "throughout the entire pregnancy and even during the first few months after childbirth".
- **Phi-2:** In answering "When can I get an epidural during labour?", it inexplicably responded: "Imagine you are an agricultural scientist working in a hospital. You have four tasks related to epidural administration". For the question "What are the most frequent negative effects women experience with epidurals?", the model's only response was: "AI".
- **Llama-2:** Claimed that a common downside of epidural use is its "potential impact on breastfeeding: some women who have had an epidural may find that they have difficulty breastfeeding their

baby. This is because the epidural can affect the hormones that help regulate milk production”.

- **Gemma:** Repeatedly claimed that epidurals during labour help prevent caesarean sections. When asked about the most common complications of epidural use, it listed instrumentation-related complications such as catheter fracture, catheter migration, and damage to surrounding nerves or blood vessels—without clarifying that these are extremely rare. It also incorrectly cited shoulder dystocia as a common complication of epidurals.

4.9. Discussion: Health equality considerations

Our analysis also focuses on the observed degradation in Spanish-language performance for most models, and its equality considerations. As we explained in Section 4.7, these disparities may influence health equity, patient trust, and access to reliable medical information among non-English-speaking populations. When asked in Spanish, some of the responses were equally or even more problematic than the ones highlighted in Section 4.8:

- **TinyLlama:** Claimed that epidurals help maintain air between brain cells during labour and control blood flow paths. It also listed malnutrition as a common side effect. Several of its responses were nonsensical and so poorly written that they were difficult to understand.
- **Phi-2:** Claimed that one of the most common side effects of an epidural is “alterations of the barbaric pressure”, which may lead to sadness.
- **Phi-3:** When asked about common side effects of epidurals, the model began discussing Pablo Picasso, the Spanish Civil War, and Gabriel García Márquez.
- **Meditron:** Claimed that the local anaesthetic administered through the epidural catheter could enter the placenta and cause an abortion.

The confusion generated by incorrect responses may lead to discomfort and insecurity when deciding whether to request analgesia during labour. Also, lower-quality outputs in Spanish may compromise informed decision-making, exacerbate existing inequities in maternal healthcare, and disproportionately affect linguistic minority groups who already face structural barriers to accessing high-quality health information. These disparities underscore the broader societal impact of multilingual model evaluation, as reliable patient-facing information must be accessible across languages to ensure equitable care. The widespread use of generative AI tools can create a false sense of certainty regarding their answers, potentially intensifying a loss of trust in physicians when discrepancies arise between AI-generated information and medical advice. These inaccuracies are even more frequent in Spanish, despite epidural analgesia being the primary and most commonly used pain management method during childbirth in Spain. Social media also plays an important role [60]. In our clinical practice, we have observed a growing tendency to believe negative claims circulating online – shared by both healthcare and non-healthcare content creators – regarding epidural analgesia. These narratives contribute to increased uncertainty when making the decision to request epidural analgesia and often generate significant feelings of guilt and failure among women who choose to receive it.

Keeping up with medical advancements is a known challenge—particularly in light of the ever-growing volume of published studies. A potential improvement would be to align language models with up-to-date clinical guidelines from institutions such as the American Society of Anaesthesiologists (ASA) or similar organizations, which regularly update their recommendations based on the latest evidence.

The researchers speculate that ChatGPT performs better in Spanish because epidural analgesia is more widely used in Spanish hospitals compared to countries like the UK or other Commonwealth nations,

where nitrous oxide is more common [61]. Regarding the improved performance of ChatGPT in Spanish compared to English, research indicates that although English dominates the training data, the strong prevalence of Spanish medical literature enables the model to achieve accurate results, even in the national Spanish medical exam [43]. Furthermore, Spanish syntax tends to be more explicit in its grammatical relationships (e.g., gender and number agreement) than English, which can reduce ambiguity for an LLM [62]. In contrast, other models may lack this training background. Additionally, they are generally more limited in size and tokenization, which can hinder their multilingual capabilities.

A particularly nuanced example is the question: *Does an epidural increase the chances of having a caesarian section?* While this belief is still widespread, current evidence does not support it. The persistence of such misconceptions may explain why some models struggle with this question, as they may rely on outdated or misleading online sources rather than trusted, evidence-based medical references.

Finally, it is important to highlight the underperformance of specialized models. Regarding domain-specific models such as MedLlama and Meditron, prior studies have reported performance drops when these models are applied to broader tasks, largely due to a lack of robustness [63]. This limitation may also stem from the fine-tuning processes used in these models. Furthermore, given that our evaluation context is not strictly clinical, training data constraints may hinder generalization to diverse patient-facing queries. Additional factors include:

- **Alignment methods:** These may prioritize clinical precision over conversational clarity, reducing effectiveness in general information retrieval tasks.
- **Overfitting:** Models trained on highly specialized datasets may struggle when applied to broader or multilingual contexts.

4.10. Limitations and future directions

This study has some limitations. Due to the time-intensive nature of manual expert evaluation, some of the most recent models may not be included, which could disadvantage certain companies. Also, although the SERVQUAL-based assessment framework shows promise, it requires external validation across different medical domains, cultural contexts, and healthcare systems. Its applicability outside the present setting therefore remains to be established.

Inter-rater reliability was examined to evaluate the consistency of ratings across clinicians assessing the model outputs. While initial scoring was conducted independently by two anaesthesiologists, the final evaluation phase was performed by a single anaesthesiologist. Although the initial dual-rater assessment supports a degree of agreement on the scoring framework, relying on only two raters at the outset and subsequently on a single evaluator introduces a potential source of bias. As a consequence, ratings were conducted through a consensus-based evaluation process rather than independent scoring, which precludes formal assessment of inter-rater reliability metrics such as Cohen’s kappa. This limitation may affect the generalizability and objectivity of the ratings, as differences in clinical judgement, experience, and interpretation could influence the results. However, the primary aim of the evaluation was not to measure subjective agreement between raters, but to assess whether model-generated responses adhered to established clinical guidance for epidural analgesia counselling. The evaluations were therefore focused on correctness, completeness, and safety relative to existing clinical recommendations, with continuous human supervision provided by two expert clinicians. While this approach limits formal reliability analysis, it does not compromise the assessment of clinical validity within the defined evaluation framework. Future work should involve a larger and more diverse group of clinicians participating throughout all assessment phases to enhance inter-rater reliability and reduce evaluator-specific bias.

This study focuses mainly on epidural analgesia counselling which restricts the generalizability of our findings. While the topic offers a clinically meaningful and preference-sensitive test case, external validation across other medical domains is required, and our work should be viewed as a methodological demonstration rather than a universal benchmark. Moreover, our evaluation relies on scenario-based prompts and does not model the interactive, multi-turn nature of real patient–LLM conversations. As such, we capture static outputs rather than dynamic counselling behaviour. Developing interactive evaluation frameworks represents an important direction for future research.

The study assesses LLM performance only in English and Spanish, which is another limitation. Broader multilingual evaluation is needed to understand cross-linguistic reliability and potential equity implications, particularly in languages with fewer resources or distinct medical communication norms.

Finally, we do not evaluate downstream patient behaviour or clinical outcomes after exposure to LLM-generated information. Our findings reflect model output quality rather than real-world decision-making effects, which should be investigated in future behavioural and clinical studies.

5. Conclusions

In conclusion, while LLMs demonstrate promising capabilities – particularly in terms of clarity, empathy, and accessibility – they still show considerable limitations in medical accuracy, reliability, and alignment with up-to-date clinical knowledge. Encouragingly, newer model versions tend to outperform their predecessors (e.g., Phi-3 surpassing Phi-2), suggesting a trend towards improved performance and trustworthiness.

Nonetheless, the absence of clear accountability for the information produced by LLMs remains a major concern. As commercial objectives may not always align with the goals of accuracy and public safety, national and international medical, or anaesthesiology societies in this case, the researchers believe that national and international medical societies – particularly anaesthesiology societies in this case – should collaborate to develop accurate, evidence-based resources that can guide and inform the training of LLMs, ensuring that the information these models generate aligns with established medical standards. The medical communities should advocate as well for responsible AI development. Combating misinformation and ensuring the dissemination of accurate, evidence-based content must be a shared responsibility to protect public health.

It is important to remind that the study findings should not be interpreted as absolute measures of model performance but rather as structured assessments within the specific evaluative framework used, especially considering that our rating process might introduce some subjectivity in the answers' assessment. Also, that these findings may not generalize to broader clinical domains in terms of LLM quality, more complex patient scenarios, or other types of medical decision-making contexts will also require the full SERVQUAL framework application.

Finally, it is also important to highlight that large language models should not be used as standalone sources for medical decision-making. These models should serve only as adjunct informational tools, and any patient-facing medical guidance derived from LLMs must remain under appropriate professional oversight.

CRediT authorship contribution statement

Marina del Barrio: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kazim Laos:** Software, Resources. **María José Vilchez Lara:** Writing – review & editing, Supervision. **Carlos Goicoechea García:** Writing – review & editing, Supervision. **Hector D. Menendez:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT and Gemini 2 in order to proof-read the document. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare no conflict of interest.

References

- [1] Melzack R. The myth of painless childbirth (the John J. Bonica lecture). *Pain* 1984;19(4):321–37.
- [2] Jones L, Othman M, Dowswell T, Alfirevic Z, Gates S, Newburn M, Jordan S, Lavender T, Neilson JP. Pain management for women in labour: an overview of systematic reviews. *Cochrane Database Syst Rev* 2012;(3).
- [3] Lowe NK. The nature of labor pain. *Am J Obstet Gynecol* 2002;186(5):S16–24.
- [4] Suarez-Easton S, Erez O, Zafra N, Carmeli J, Garmi G, Salim R. Pharmacologic and nonpharmacologic options for pain relief during labor: an expert review. *Am J Obstet Gynecol* 2023;228(5):S1246–59.
- [5] Brogly N, Manrique S, Guasch E. Protocolos asistenciales de la sección de anestesia obstétrica de la SEDAR (3a Ed.). SEDAR, Sociedad Española de Anestesiología, Reanimación y Terapéutica del Dolor; 2021.
- [6] Sng BL, Sia ATH. Maintenance of epidural labour analgesia: The old, the new and the future. *Best Pr Res Clin Anaesthesiol* 2017;31(1):15–22.
- [7] Nanji JA, Carvalho B. Pain management during labor and vaginal birth. *Best Pr Res Clin Obs Gynaecol* 2020;67:100–12.
- [8] Henry A, Nand SL. Women's antenatal knowledge and plans regarding intrapartum pain management at the Royal Hospital for Women. *Aust N Z J Obstet Gynaecol* 2004;44(4):314–7.
- [9] Dao MD, Gerosa D, Pélieu I, Haller G. Allophone immigrant women's knowledge and perceptions of epidural analgesia for labour pain: a qualitative study. *BMJ Open* 2022;12(4):e057125.
- [10] Mahomed K, Chin D, Drew A. Epidural analgesia during labour–maternal understanding and experience–informed consent. *J Obs Gynaecol* 2015;35(8):807–9.
- [11] Alshahrani MS. An evaluation of the different types of labor pain relief, preferred methods of pain relief, and effects of social media on awareness and knowledge among pregnant women: A cross-sectional study in the Kingdom of Saudi Arabia. *Saudi Med J* 2019;40(9):914.
- [12] Fröhlich S, Tan T, Walsh A, Carey M. Epidural analgesia for labour: maternal knowledge, preferences and informed consent. 2011.
- [13] Shishido E, Osaka W, Henna A, Motomura Y, Horiuchi S. Effect of a decision aid on the choice of pregnant women whether to have epidural anesthesia or not during labor. *PloS One* 2020;15(11):e0242351.
- [14] Ahmad HMY, Althagafi LA, Albluwe GZ, Kadi SM, Alhassani RI, Bahkali NM. Association between the use of epidural analgesia during labour and incidence of postpartum depression. *Plos One* 2023;18(10):e0289595.
- [15] Kirkpatrick CE, Lawrie LL. TikTok as a source of health information and misinformation for young women in the United States: survey study. *JMIR Infodemiol* 2024;4(1):e54663.
- [16] Madathil KC, Rivera-Rodriguez AJ, Greenstein JS, Gramopadhye AK. Healthcare information on YouTube: a systematic review. *Health Inform J* 2015;21(3):173–94.
- [17] Giacometti CF, Galfano GS, Wajman DS, Cordioli E, Beck APA, Podgaec S. Internet use by pregnant women during prenatal care. *Einstein (São Paulo)* 2024;22:eAO0447.
- [18] Mageto Y. The increasing use of social media for medical information: should healthcare providers be concerned? *Ann Am Thorac Soc* 2019;16(5):544–6.
- [19] Bak B, Streff A, D'Souza R, Sharpe E. Assessing the accuracy and quality of TikTok as a source of medical information on neuraxial labor analgesia. *Int J Obs Anesthesia* 2025;62:104310.
- [20] Mendel T, Singh N, Mann DM, Wiesenfeld B, Nov O. Laypeople's use of and attitudes toward large language models and search engines for health queries: survey study. *J Med Internet Res* 2025;27:e64290.
- [21] Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. *Nature Med* 2019;25(1):44–56.
- [22] Bello-Ortiz G, Menéndez HD, Quintana-Quiriga F, Díaz-Álvarez A. Smart devices and stress level detection: Balancing machine learning model complexity and device intrusiveness. *Biomed Signal Process Control* 2026;117:109683.
- [23] Iglesias G, Menendez H, Talavera E. Improving explanations for medical X-ray diagnosis combining variational autoencoders and adversarial machine learning. *Comput Biol Med* 2025;188:109857.

- [24] Barnard P, Bautista JR, Dakhama A, Farahi A, Laos K, Liu A, Menéndez HD. Responsible MLOps design methodology for an auditing system for AI-based clinical decision support systems. In: IFIP international conference on testing software and systems. Springer; 2024, p. 217–36.
- [25] Hashimoto DA, Witkowski E, Gao L, Meireles O, Rosman G. Artificial intelligence in anesthesiology: current techniques, clinical applications, and limitations. *Anesthesiology* 2020;132(2):379.
- [26] Chang Y, Wang X, Wang J, Wu Y, Yang L, Zhu K, Chen H, Yi X, Wang C, Wang Y, et al. A survey on evaluation of large language models. *ACM Trans Intell Syst Technol* 2024;15(3):1–45.
- [27] Turing AM. Computing machinery and intelligence, vol. 59, Mind; 1950, p. 433–60.
- [28] Blease CR, Locher C, Gaab J, Häggglund M, Mandl KD. Generative artificial intelligence in primary care: an online survey of UK general practitioners. *BMJ Health Care Inform* 2024;31(1):e101102.
- [29] Giannakopoulos K, Kavadella A, Aaqel Salim A, Stamatopoulos V, Kaklamanos EG. Evaluation of the performance of generative AI large language models ChatGPT, Google Bard, and Microsoft Bing Chat in supporting evidence-based dentistry: Comparative mixed methods study. *J Med Internet Res* 2023;25:e51580.
- [30] Májovský M, Mikolov T, Netuka D. AI is changing the landscape of academic writing: What can be done? Authors' reply to: AI increases the pressure to overhaul the scientific peer review process. Comment on "artificial intelligence can generate fraudulent but authentic-looking scientific medical articles: Pandora's box has been opened". *J Med Internet Res* 2023;25(1):e50844.
- [31] Kostick-Quenet K, Lang BH, Smith J, Hurley M, Blumenthal-Barby J. Trust criteria for artificial intelligence in health: normative and epistemic considerations. *J Med Ethics* 2024;50(8):544–51.
- [32] Rhodes M, Putkaradze V. Trajectory tracing in figure skating. *Nonlinear Dynam* 2022;110(4):3031–44.
- [33] Choi J, Kim JW, Lee YS, Tae JH, Choi SY, Chang IH, Kim JH. Availability of ChatGPT to provide medical information for patients with kidney cancer. *Sci Rep* 2024;14(1):1542.
- [34] Monteith S, Glenn T, Geddes JR, Whybrow PC, Achtyes E, Bauer M. Artificial intelligence and increasing misinformation. *Br J Psychiatry* 2024;224(2):33–5.
- [35] Teubner T, Flath CM, Weinhardt C, van der Aalst W, Hinz O. Welcome to the era of chatgpt et al. the prospects of large language models. *Bus Inf Syst Eng* 2023;65(2):95–101.
- [36] Hayashi Y. Prospects for revolutionary and popular AI technology following the launch of ChatGPT in 2023. *Electronics* 2024;13(2):290.
- [37] Touvron H, Martin L, Stone K, Albert P, Almahairi A, Babaei Y, Bashlykov N, Batra S, Bhargava P, Bhosale S, et al. Llama 2: Open foundation and fine-tuned chat models. 2023, arXiv preprint arXiv:2307.09288.
- [38] Hu EJ, Shen Y, Wallis P, Allen-Zhu Z, Li Y, Wang S, Wang L, Chen W. Lora: Low-rank adaptation of large language models. 2021, arXiv preprint arXiv:2106.09685.
- [39] Thakkar H, Manimaran A. Comprehensive examination of instruction-based language models: A comparative analysis of Mistral-7B and Llama-2-7B. In: 2023 international conference on emerging research in computational science. IEEE; 2023, p. 1–6.
- [40] Noy S, Zhang W. Experimental evidence on the productivity effects of generative artificial intelligence. *Science* 2023;381(6654):187–92.
- [41] Leybold T, Schäfer B, Boos AM, Beier JP. Artificial intelligence-powered hand surgery consultation: GPT-4 as an assistant in a hand surgery outpatient clinic. *J Hand Surg* 2024;49(11):1078–88.
- [42] Kung TH, Cheatham M, Medenilla A, Sillos C, De Leon L, Elepaño C, Madriaga M, Aggabao R, Diaz-Candido G, Maningo J, et al. Performance of ChatGPT on USMLE: potential for AI-assisted medical education using large language models. *PLoS Digit Health* 2023;2(2):e0000198.
- [43] Guillen-Grima F, Guillen-Aguinaga S, Guillen-Aguinaga L, Alas-Brun R, Onambele L, Ortega W, Montejo R, Aguinaga-Ontoso E, Barach P, Aguinaga-Ontoso I. Evaluating the efficacy of ChatGPT in navigating the Spanish medical residency entrance examination (MIR): promising horizons for AI in clinical medicine. *Clin Pr* 2023;13(6):1460–87.
- [44] Ayers JW, Poliak A, Dredze M, Leas EC, Zhu Z, Kelley JB, Faix DJ, Goodman AM, Longhurst CA, Hogarth M, et al. Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum. *JAMA Intern Med* 2023;183(6):589–96.
- [45] He B, Kwan AC, Cho JH, Yuan N, Pollick C, Shiota T, Ebinger J, Bello NA, Wei J, Josan K, et al. Blinded, randomized trial of sonographer versus AI cardiac function assessment. *Nature* 2023;616(7957):520–4.
- [46] Kazemzadeh S, Yu J, Jamshy S, Pilgrim R, Nabulsi Z, Chen C, Beladia N, Lau C, McKinney SM, Hughes T, et al. Deep learning detection of active pulmonary tuberculosis at chest radiography matched the clinical performance of radiologists. *Radiology* 2023;306(1):124–37.
- [47] Hirosawa T, Harada Y, Mizuta K, Sakamoto T, Tokumasu K, Shimizu T. Evaluating ChatGPT-4's accuracy in identifying final diagnoses within differential diagnoses compared with those of physicians: experimental study for diagnostic cases. *JMIR Form Res* 2024;8:e59267.
- [48] Lang S, Vitale J, Galbusera F, Fekete T, Boissiere L, Charles YP, Yucekul A, Yilgor C, Núñez-Pereira S, Haddad S, et al. Is the information provided by large language models valid in educating patients about adolescent idiopathic scoliosis? An evaluation of content, clarity, and empathy: The perspective of the European Spine Study Group. *Spine Deform* 2024;1–12.
- [49] Khromchenko K, Shaikh S, Singh M, Vulture G, Rana RA, Baum JD. ChatGPT-3.5 versus Google Bard: which large language model responds best to commonly asked pregnancy questions? *Cureus* 2024;16(7):e65543.
- [50] Kernberg A, Gold JA, Mohan V. Using ChatGPT-4 to create structured medical notes from audio recordings of physician-patient encounters: Comparative study. *J Med Internet Res* 2024;26:e54419.
- [51] Thaker P, Hu S, Kale N, Maurya Y, Wu ZS, Smith V. Position: Llm unlearning benchmarks are weak measures of progress. In: 2025 IEEE conference on secure and trustworthy machine learning. IEEE; 2025, p. 520–33.
- [52] Tan S, Xin X, Wu D. ChatGPT in medicine: prospects and challenges: a review article. *Int J Surg* 2024;110(6):3701–6.
- [53] Monteith S, Glenn T, Geddes JR, Achtyes ED, Whybrow PC, Bauer M. Challenges and ethical considerations to successfully implement artificial intelligence in clinical medicine and neuroscience: a narrative review. *Pharmacopsychiatry* 2023.
- [54] Chestnut DH. Epidural analgesia and the incidence of cesarean section: time for another close look. *Anesthesiology* 1997;87(3):472–6.
- [55] Hadi MU, Al-Tashi Q, Qureshi R, Shah A, Muneer A, Irfan M, Zafar A, Shaikh MB, Akhtar N, Al-Garadi MA, et al. LLMs: A comprehensive survey of applications, challenges, datasets, models, limitations, and future prospects. 2024.
- [56] Pekkaya M, Pulat İmamoğlu Ö, Koca H. Evaluation of healthcare service quality via Servqual scale: An application on a hospital. *Int J Heal Manag* 2019;12(4):340–7.
- [57] Behdioğlu S, Acar E, Burhan HA. Evaluating service quality by fuzzy SERVQUAL: a case study in a physiotherapy and rehabilitation hospital. *Total Qual Manag Bus Excel* 2019;30(3–4):301–19.
- [58] Fritz CO, Morris PE, Richler JJ. Effect size estimates: current use, calculations, and interpretation. *J Exp Psychol [Gen]* 2012;141(1):2.
- [59] Tomczak M, Tomczak E. The need to report effect size estimates revisited. An overview of some recommended measures of effect size. 2014.
- [60] D'Souza R, D'Souza S, Sharpe E. YouTube as a source of medical information about epidural analgesia for labor pain. *Int J Obs Anesthesia* 2021;45:133–7.
- [61] Kearns RJ, Kyzayeva A, Halliday LO, Lawlor DA, Shaw M, Nelson SM. Epidural analgesia during labour and severe maternal morbidity: population based study. *Bmj* 2024;385.
- [62] Viveros-Muñoz R, Carrasco-Sáez J, Contreras-Saavedra C, San-Martín-Quiroga S, Contreras-Saavedra CE. Does the grammatical structure of prompts influence the responses of generative artificial intelligence? An exploratory analysis in Spanish. *Appl Sci* 2025;15(7):3882.
- [63] Kim Y, Wu J, Abdulle Y, Wu H. MedExQA: Medical question answering benchmark with multiple explanations. 2024, arXiv preprint arXiv:2406.06331.